

Symmetry Alignment Across Depth in Transformer Merging

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Abstract

Symmetry-based alignment can yield zero-barrier paths between independently trained Transformers at fixed depth. We test whether this picture extends across depth by padding the shallower model with exact identity blocks. The padded model shares the deep model's L -block parameter space but lies in an invariant subset where selected residual branches vanish. Across five depth pairs on WikiText-2, Penn Treebank, and Tiny Shakespeare, inherited permutation-and-orthogonal (P+O) alignment removes 90.1–94.6% of the naive barrier yet leaves a positive residual of 0.098–0.244 nats in every tested configuration. Additional width ($d \in \{256, 512, 768\}$ where available) and layer-placement ablations show that the residual depends strongly on representation width and on where identity blocks are inserted. Raw identity-slot count does not order the residuals, and matched-ratio depth trends are corpus-dependent rather than universal.

Keywords: linear mode connectivity, parameter space symmetry, quotient geometry, model merging, Transformers, orthogonal group, loss landscape geometry

1. Introduction

Symmetry-aware alignment can turn high-loss interpolation between independently trained networks into low- or zero-barrier paths by quotienting function-preserving reparameterizations. For Transformers, Theus et al. (2025) extend this beyond permutations and recover zero-barrier paths at matched depth. We ask whether that fixed-depth picture survives depth mismatch. We insert exact identity residual blocks into the shallower model and apply the inherited alignment class. Padding fixes shape but leaves the expanded model in an invariant zero-branch subset that a generic deep model does not occupy. Across three language corpora, the alignment remains highly effective but consistently leaves a positive residual. We study this as a boundary case of fixed-depth symmetry alignment, not as an irreducibility theorem.

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2. Related Work and Positioning

Permutation-aware views of linear mode connectivity motivate quotienting hidden-unit permutations (Entezari et al., 2022); Git Re-Basin makes this constructive through explicit permutation alignment (Ainsworth et al., 2023), while optimal-transport fusion provides another alignment-based route to model merging (Singh and Jaggi, 2020). Residual barriers need not diagnose missing symmetries alone: REPAIR shows that interpolation pathologies can persist after alignment and can be corrected post hoc (Jordan et al., 2023). Most directly, Theus et al. (2025) enlarge Transformer reparameterizations beyond permutations and obtain low- or zero-barrier paths for independently trained models, including width-heterogeneous settings. Depth heterogeneity has instead been addressed algorithmically by Xu et al. (2024), who merge groups of deep layers with shallower layers. Our question is deliberately narrower and geometric: what remains of the fixed-depth symmetry picture after the minimal function-preserving embedding that makes unequal-depth graphs shape compatible?

3. Setup

Models and the alignment class. We use pre-norm GPT-2 decoders (Radford et al., 2019) with parameter-free RMSNorm (Zhang and Sennrich, 2019) and tied embeddings. Deep and shallow models have L and $\ell < L$ blocks and otherwise matched architecture settings. Following Theus et al. (2025), our tested P+O class combines block-wise permutations with a global orthogonal residual-stream map $O \in \mathcal{O}(d)$; call its function-preserving transformations $\mathcal{G}_L$. Its block-wise factors presuppose equal depth, so it does not directly act on an (ℓ, L) pair.

Identity-block expansion. Let ϕ^{id} have identically zero residual branches (all branch parameters, including biases if present, are zero), hence $f^{\text{id}}(x) = x$. For an injective $m : [\ell] \rightarrow [L]$, let $F = m([\ell])$, $I = [L] \setminus F$, and let $\mathcal{E}_m$ place shallow blocks in F and identities in I . Then $f_{\mathcal{E}_m(\theta^S)} = f_{\theta^S}$. Writing $\tilde{\theta}^S = \mathcal{E}_m(\theta^S)$ puts both endpoints in the same L -block space. Unless otherwise stated, we use the even-spread map $m(j) = \lfloor jL/\ell \rfloor$.

Barrier and alignment. For $T \in \mathcal{G}_L$ we interpolate $\theta(\lambda) = (1 - \lambda)\theta^D + \lambda T(\tilde{\theta}^S)$ and measure excess loss over the linear baseline,

$$\Delta(\lambda) = \mathcal{L}(\theta(\lambda)) - [(1 - \lambda)\mathcal{L}(\theta^D) + \lambda\mathcal{L}(\theta^S)]. \quad (1)$$

We evaluate $\lambda \in \{0, 0.1, \dots, 1\}$ and report the midpoint $\bar{\Delta} = \Delta(0.5)$; $\bar{\Delta}^*$ denotes the best residual found after alignment. We compare vanilla, activation and data-free weight matching (Ainsworth et al., 2023; Schönemann, 1966), learned permutations (Cuturi, 2013; Mena et al., 2018), and jointly learned P+O; optimization details are in Appendix A.

4. What Padding Changes Geometrically

Let $r_i(x; \theta_i) = f_i(x; \theta_i) - x$ be the aggregate residual update of block i , and define

$$\mathcal{Z}_I = \{\theta : r_i(\cdot; \theta_i) \equiv 0 \text{ for all } i \in I\}. \quad (2)$$

Table 1: Full P+O residual barrier $\bar{\Delta}^*$ for the depth sweep ($d = 512$, even-spread; mean $\pm$ SE over three seed pairs). Parentheses give reduction relative to vanilla averaging.

(ℓ, L)	ρ	WikiText-2	PTB	Tiny Shakespeare
(3, 6)	2.00	0.109 $\pm$ 0.001 (94.6%)	0.191 $\pm$ 0.002 (91.5%)	0.117 $\pm$ 0.007 (92.1%)
(6, 12)	2.00	0.138 $\pm$ 0.002 (94.1%)	0.223 $\pm$ 0.004 (90.8%)	0.106 $\pm$ 0.005 (92.7%)
(8, 12)	1.50	0.148 $\pm$ 0.001 (93.5%)	0.232 $\pm$ 0.001 (90.1%)	0.110 $\pm$ 0.006 (92.2%)
(6, 18)	3.00	0.134 $\pm$ 0.002 (93.9%)	0.216 $\pm$ 0.005 (90.3%)	0.106 $\pm$ 0.003 (92.5%)
(12, 24)	2.00	0.161 $\pm$ 0.001 (93.2%)	0.244 $\pm$ 0.004 (90.3%)	0.098 $\pm$ 0.003 (91.9%)

By construction $\tilde{\theta}^S \in \mathcal{Z}_I$. P+O transformations preserve zero branch parameters, so $\mathcal{Z}_I$ is invariant and $\mathcal{G}_L \cdot \tilde{\theta}^S \subseteq \mathcal{Z}_I$, whereas a generic deep endpoint with nonzero branches on I lies outside it. Padding therefore removes an ambient-space mismatch without making the endpoints symmetry equivalent. This does *not* imply a positive loss barrier: an interpolated Transformer block is nonlinear in its parameters, depends on preceding aligned blocks, and can be compensated downstream. We make no lower-bound or irreducibility claim.

5. Experiments

We evaluate the same depth-heterogeneous alignment protocol on WikiText-2, Penn Treebank (PTB), and Tiny Shakespeare. The main sweep uses five depth pairs at $d = 512$ with the even-spread map and three seed pairs per configuration. We additionally vary width on (6, 12) and compare four layer-placement strategies on PTB and Tiny Shakespeare.

5.1. Cross-Corpus Robustness

Table 1 gives the central empirical result. Across all 15 depth-pair-corpus combinations, full P+O removes 90.1–94.6% of the naive barrier, yet a positive residual of 0.098–0.244 nats remains. The phenomenon therefore survives substantial changes in corpus statistics and loss scale, while remaining small relative to vanilla interpolation. We call this a systematic residual under the tested protocol, not an unavoidable cross-depth floor.

5.2. What Controls the Residual Magnitude

Raw identity-slot count does not order the residual. On WikiText-2, (6, 18) has twelve identity slots but a smaller barrier than (8, 12) with four (0.134 versus 0.148); PTB shows the same reversal (0.216 versus 0.232), and Tiny Shakespeare is likewise non-monotone. Nor is absolute depth a universal explanation: the matched-ratio sequence $(3, 6) \rightarrow (6, 12) \rightarrow (12, 24)$ rises on WikiText-2 ($0.109 \rightarrow 0.138 \rightarrow 0.161$) and PTB ($0.191 \rightarrow 0.223 \rightarrow 0.244$), but decreases on Tiny Shakespeare ($0.117 \rightarrow 0.106 \rightarrow 0.098$).

Width and placement matter more consistently. At (6, 12), the residual grows with available width comparisons: PTB gives $0.176 \rightarrow 0.223 \rightarrow 0.254$ and Tiny Shakespeare gives $0.064 \rightarrow 0.106 \rightarrow 0.168$ for $d = 256, 512, 768$; WikiText-2 increases from 0.116 to 0.138

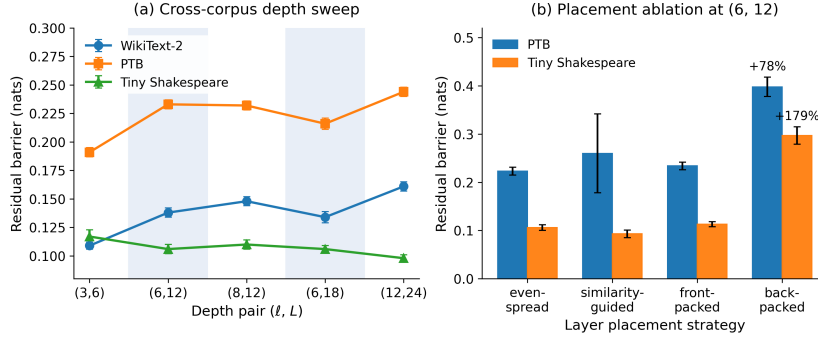


Figure 1: Cross-corpus structure of the residual barrier. Left: depth-sweep behavior across WikiText-2, PTB, and Tiny Shakespeare; matched-ratio depth trends are not universal across corpora. Right: layer-placement ablation on (6, 12) for PTB and Tiny Shakespeare, showing that early identity placement (**back-packed**) substantially increases the residual.

between $d = 256$ and 512. Layer placement is even more consequential. Relative to even-spread, **back-packed** placement raises the residual from 0.223 to 0.397 on PTB (+78%) and from 0.106 to 0.296 on Tiny Shakespeare (+179%). This shows that where identity slots are inserted strongly affects the barrier, while no single alternative mapping is uniformly best across corpora.

6. Discussion

Identity expansion exposes a distinction between *ambient-space* compatibility and *orbit* compatibility. It embeds an ℓ -block model into the same L -block parameterization as the deep model, and inherited symmetry alignment remains highly effective across three corpora. But the padded endpoint stays inside the invariant zero-branch set $\mathcal{Z}_I$, so the fixed-depth symmetry class does not identify a generic deep block with an inserted identity slot. A genuinely cross-depth construction could instead enlarge the equivalence relation to include identity insertion/deletion, or avoid slot-wise correspondence altogether through segment-based heterogeneous merging (Xu et al., 2024).

The new cross-corpus results strengthen the claim that a residual is a reproducible boundary of the tested alignment protocol, while weakening any simple scaling explanation based only on slot count or absolute depth. Its magnitude varies with corpus, width, and especially layer placement. We still do not establish irreducibility: the residual may reflect the restricted P+O class, interpolation pathology, or imperfect optimization. Remaining limitations include one architecture family, three seeds, no matched-depth rerun under our exact implementation, no systematic alignment-restart study, and no $d = 768$ WikiText-2 point; the PTB similarity-guided placement estimate also has substantial variance.

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Appendix A. Additional Experimental Detail

Alignment. Activation matching solves a Hungarian assignment on the cross-correlation of residual stream, MLP-hidden and attention activations. Weight matching alternates a Git

Re-Basin permutation phase with a closed-form orthogonal Procrustes step. The permutation learner uses a Sinkhorn relaxation with Hungarian discretization and a straight-through estimator (Bengio et al., 2013). Learned variants optimize the interpolation window around the midpoint. Unless a placement ablation is specified, runs use the even-spread layer map; for $(\ell, L) = (8, 12)$ this gives $F = \{0, 1, 3, 4, 6, 7, 9, 10\}$.

Table 2: Layer-placement ablation on $(6, 12)$, $d = 512$ (mean $\pm$ SE over three seed pairs).

Placement	Penn Treebank		Tiny Shakespeare	
	Vanilla	Full P+O	Vanilla	Full P+O
even-spread	2.422 ± 0.035	0.223 ± 0.004	1.451 ± 0.028	0.106 ± 0.005
similarity-guided	1.964 ± 0.055	0.260 ± 0.082	1.734 ± 0.044	0.093 ± 0.006
front-packed	2.616 ± 0.042	0.234 ± 0.005	1.432 ± 0.031	0.114 ± 0.002
back-packed	2.772 ± 0.061	0.397 ± 0.017	1.872 ± 0.050	0.296 ± 0.020

Table 3: Width generalization on $(6, 12)$ with even-spread placement (mean $\pm$ SE over three seed pairs). Parentheses give vanilla-barrier reduction.

d	WikiText-2	Penn Treebank	Tiny Shakespeare
256	0.116 ± 0.002 (94.1%)	0.176 ± 0.004 (91.4%)	0.064 ± 0.003 (96.0%)
512	0.138 ± 0.002 (94.1%)	0.223 ± 0.004 (90.8%)	0.106 ± 0.005 (92.7%)
768	n/a	0.254 ± 0.005 (90.8%)	0.168 ± 0.008 (89.2%)

Table 4: Selected method-level results from the multi-corpus evaluation (mean $\pm$ SE over three seed pairs).

Dataset	Configuration	Vanilla	Weight	Full P+O
WikiText-2	$(3, 6), d = 512$	2.031 ± 0.046	0.377 ± 0.023	0.109 ± 0.001
WikiText-2	$(6, 12), d = 512$	2.358 ± 0.049	0.459 ± 0.012	0.138 ± 0.002
WikiText-2	$(8, 12), d = 512$	2.341 ± 0.075	0.518 ± 0.025	0.148 ± 0.001
WikiText-2	$(6, 18), d = 512$	2.182 ± 0.022	0.471 ± 0.014	0.134 ± 0.002
WikiText-2	$(12, 24), d = 512$	2.376 ± 0.125	0.514 ± 0.012	0.161 ± 0.001
PTB	$(6, 12), d = 512$	2.422 ± 0.035	0.512 ± 0.021	0.223 ± 0.004
Tiny Shakespeare	$(6, 12), d = 512$	1.451 ± 0.028	0.205 ± 0.012	0.106 ± 0.005

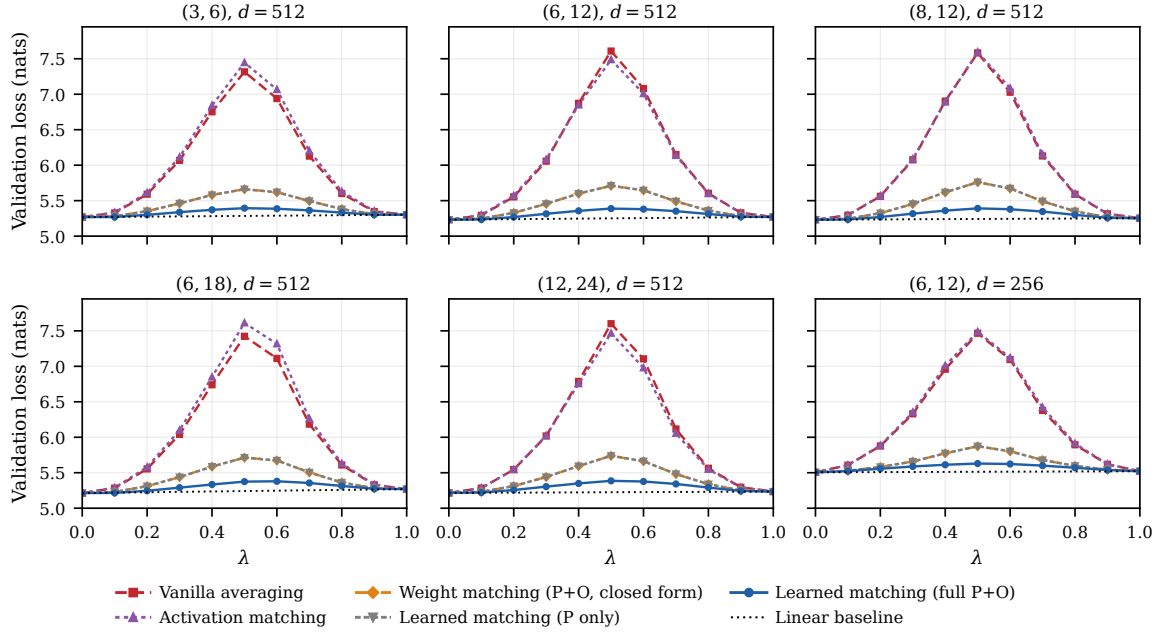


Figure 2: Interpolation paths for the WikiText-2 configurations, mean over three seed pairs, on a shared axis. The ordering of methods is identical throughout. Because the shared vertical scale includes the much larger vanilla barriers, the remaining Full P+O excess is visually compressed; the main text reports its magnitude directly.

